

| Ptychography, Tomography                                                                                                                                      | MUMPS                                                                                                                          | XK.jl, Krylov.jl                                                                                                                   | OpenMP Applications                                                                                                                                                   |
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| <b>Drava Runtime</b><br>Event-Driven Programming Model<br>Real-time high-throughput data processing<br>C, C++ and Python API                                  | <b>XKBlas Library</b><br>Partial support for BLAS 1, 2 and 3<br>Task-spawning algorithms to multi-GPUs<br>C, C++ and Julia API |                                                                                                                                    | <b>XKOMP Abstraction Layer</b><br>Pragma API via GOMP, KMP and libomptarget ABI<br>Custom XKOMP ABI (functionality, performance)<br>C++ API (no pragmas) with lambdas |
| <b>XKRT Parallel Runtime System</b><br>Tasks, threads, teams, NUMA awarness, workstealing, etc.<br>Device management (initialization, memory, commands, etc.) |                                                                                                                                |                                                                                                                                    |                                                                                                                                                                       |
| <b>Hardware Vendors APIs</b><br>CUDA, HIP, Level Zero, OpenCL<br>compatible-devices (e.g., CPU, GPU, FPGA)                                                    |                                                                                                                                | <b>OpenCG Library</b><br>An IR for optimizing graph of commands on<br>multi-devices systems. Custom POD IR and MLIR JIT with LLVM. |                                                                                                                                                                       |